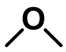
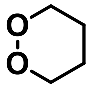
CHM102 Basic Organic Chemistry

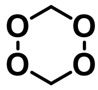
Problem Set 1

Q1. Identify the molecule with higher dipole moment and indicate the reason.

 vs 

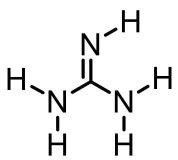
Q2. Two cyclic peroxides (A and B) are shown. (i) Identify the IUPAC names of these compounds. (ii) Among these, the bis-peroxide B is more stable than monoperoxide A. Why?


A

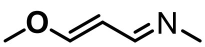
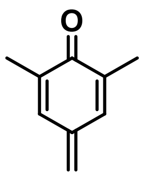
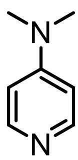
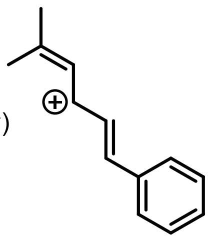
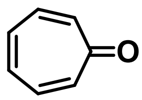

B

Q3. Draw the frontier molecular orbitals of (i) benzene, (ii) 1,3,5,7-octatetraene and (iii) cyclobutadiene.

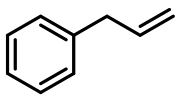
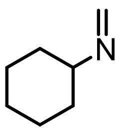
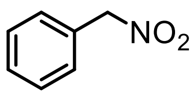
Q4. The structure of guanidine shows that the C-N bond distances are unequal (1.295, 1.369 and 1.368 Å). However, the protonation of it makes all the three C-N bond distances equal. Indicate the reason.



Q5. Draw all the possible resonance structures of the following, and indicate the most stable canonical structure among the resonance structures:

(i)  (ii)  (iii)  (iv)  (v) 

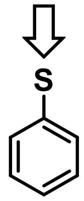
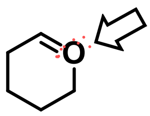
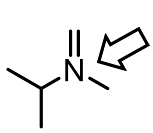
Q6. Draw the possible tautomeric structures for the following compounds with the equilibrium indicating the most stable structure. Also, indicate the reason.

(a)  (b)  (c) 

Q7. Draw the following MOs

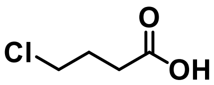
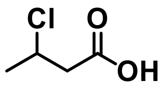
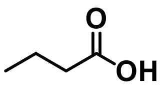
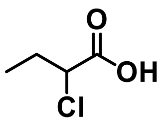
(a) LUMO of 1,3-butadiene (b) LUMO of allyl cation (c) LUMO of 1,3,5-hexatriene.

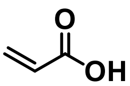
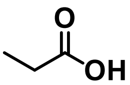
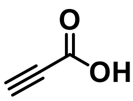
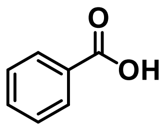
Q8. Estimate the formal charge at the indicated atom within the given molecules.

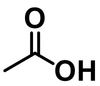
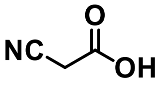
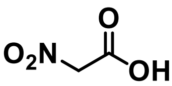
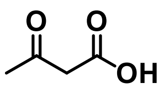
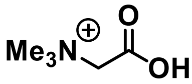
(i)  (ii)  (iii) 

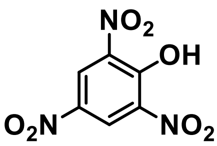
Q9. In indole, the electrophilic substitution happens at 3-position rather than 2-position, which is in contrast to pyrrole. Why?

Q10. Among the following sets of acids, arrange them in terms of increasing acidity and indicate the reason for the variation in the acidity.

(i)  vs  vs  vs 

(ii)  vs  vs  vs 

(iii)  vs  vs  vs  vs 

(iv)  vs